

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA1003	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	ALIGETI AKSHAYA	Reg. No.:	192425170
Date:	24-08-2026	Duration:	60 minutes

Code of Conduct

I A.Akshaya(192425170)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____Akshaya_____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

**ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION
MANAGEMENT SYSTEM**

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
 - ☐ Manlove & O'Malley (2008) – Student-Project Allocation with Preferences over Projects
 - ☐ Abraham, Irving & Manlove (2007) – Two Algorithms for the Student-Project Allocation Problem
 - ☐ Chiarandini, Fagerberg & Gualandi (2019) – Handling Preferences in Student-Project Allocation
 - ☐ Manlove, Milne & Olaosebikan (2022) – Student-Project Allocation with Preferences over Projects: Algorithmic and Experimental Results

□ *Hoang et al. (2025)* – An Efficient Two-Heuristic Algorithm for the Student-Project Allocation with Preferences over Projects

□ Dynamic Matching Strategy for College Students' Innovative Training Projects Based on Reinforcement Learning Optimization (2026)

2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.

Paper 1: Manlove & O'Malley – 2008

Title: *Student-Project Allocation with Preferences over Projects*

Authors: David F. Manlove and Gregg O'Malley

Journal: Journal of Discrete Algorithms, 2008.

Objective

The paper studies how students can be allocated to projects while considering **student preferences, lecturer preferences, and capacity constraints**.

Methodology

The researchers formulate the problem as a **stable matching problem**. Students have preferences over projects, while lecturers and projects have capacity limitations.

Technologies / Algorithm

- Stable matching
- Preference lists
- Matching algorithms
- Approximation algorithms

Dataset

The work is primarily **algorithmic/theoretical**, rather than being based on a machine-learning dataset.

Major Findings

The researchers show that finding a maximum-size stable matching is computationally difficult in the general case. They also provide an approximation algorithm with a performance guarantee of 2.

Limitation

The model mainly concentrates on **preferences and capacities**. It does not consider detailed student technical skills, project progress, evaluation, or feedback.

Paper 2: Abraham, Irving & Manlove – 2007

Title: *Two Algorithms for the Student-Project Allocation Problem*

Journal: Journal of Discrete Algorithms, 2007.

Objective

The objective is to develop efficient algorithms for assigning students to projects while satisfying preferences and capacity constraints.

Methodology

The paper proposes **two linear-time algorithms** for the Student-Project Allocation problem.

One produces a student-optimal stable matching, while another produces a lecturer-optimal stable matching.

Technologies / Algorithms

- Stable matching
- Preference lists
- Linear-time algorithms
- Capacity constraints

Dataset

The paper is algorithm-focused rather than dependent on a conventional dataset.

Major Findings

The algorithms can produce stable allocations efficiently. The paper also demonstrates that stable matching can be useful for applications involving students, projects, and lecturers.

Limitation

The approach primarily considers **preferences and capacity** rather than a comprehensive student profile containing skills, interests, previous knowledge, and learning requirements.

Paper 3: Chiarandini, Fagerberg & Gualandi – 2019

Title: *Handling Preferences in Student-Project Allocation*

Journal: Annals of Operations Research, 2019.

Objective

The research aims to allocate students to projects while considering **preferences, fairness, stability, and side constraints**.

Methodology

The problem is formulated as a **Mixed Integer Linear Programming (MILP)** problem.

Different models are used to balance:

- Student satisfaction
- Fairness
- Stability
- Collective satisfaction

Dataset / Data

The research uses **real-world data from the Faculty of Science at the University of Southern Denmark** and evaluates different allocation models using a commercial optimization solver.

Major Findings

The researchers found combinations of models that could produce **feasible, stable, fair, and collectively satisfactory allocations within about a minute** for their institutional setting.

Limitation

The focus is mainly on **allocation optimization**. It does not provide an integrated platform for continuous project monitoring, evaluation, feedback, and student skill development.

Paper 4: Manlove, Milne & Olaosebikan – 2022

Title: *Student-Project Allocation with Preferences over Projects: Algorithmic and Experimental Results*

Journal: Discrete Applied Mathematics, 2022.

Objective

The paper evaluates algorithms for the Student-Project Allocation problem and compares their results with optimal solutions.

Methodology

The researchers use:

- Approximation algorithms
- Integer Programming
- Empirical evaluation

They compare algorithmic solutions with optimal solutions obtained through an **Integer Programming model**.

Dataset

Experiments use both:

- Randomly generated instances
- Instances derived from real datasets

Major Findings

The research demonstrates that maximum-size stable allocation can be computationally difficult. The Integer Programming model can obtain optimal solutions for general instances, while approximation algorithms provide practical alternatives.

Limitation

The study is focused on the **allocation algorithm itself**, rather than a complete academic project management system.

Paper 5: Hoang et al. – 2025

Title: *An Efficient Two-Heuristic Algorithm for the Student-Project Allocation with Preferences over Projects*

Journal: Journal of Intelligent & Fuzzy Systems, 2025.

Objective

The research aims to develop a more efficient approach for the Student-Project Allocation problem.

Methodology

The researchers propose a **two-heuristic algorithm** that iteratively constructs a stable matching by assigning students to preferred projects while considering constraints.

Technology / Algorithm

- Heuristic algorithms
- Stable matching
- Preference-based allocation
- Optimization

Major Findings

The proposed approach focuses on efficiently constructing maximum stable allocations for student-project matching problems.

Limitation

The approach is still mainly concerned with **matching students with projects**. It does not cover the complete lifecycle of an academic project.

Paper 6: Dynamic Matching Strategy – 2026

Title: *Dynamic Matching Strategy for College Students' Innovative Training Projects Based on Reinforcement Learning Optimization*

Published: 2026, *Discover Artificial Intelligence*.

Objective

This research addresses the problem of traditional project allocation methods that may overlook students' **individual interests, skills, and learning styles**.

Methodology

The researchers propose a **dynamic matching algorithm based on Reinforcement Learning (RL)**.

The system analyzes:

- Student profiles
- Interests
- Skill ratings
- Project metadata
- Interactions between students and projects

The allocation process is continuously optimized using reinforcement learning.

Major Finding

The research moves beyond static allocation and proposes a **dynamic intelligent matching approach**, which is particularly close to the idea behind your project.

Limitation

Even though it considers interests and skills, the research focuses mainly on **matching students with projects** rather than providing an integrated platform covering allocation, faculty monitoring, progress tracking, evaluation, feedback, and reporting.

3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Manlove & O'Malley, 2008	Student-project allocation with preferences and capacity	Stable matching, approximation	Algorithmic instances	Stable allocation can be generated while respecting preferences/capacities	Mainly preferences and capacities	Does not consider student skills, progress or evaluation
Abraham et al., 2007	Efficient student-project	Two linear-time stable matching	Algorithmic problem instance	Produces student-optimal and lecturer-optimal	Limited student information	No integrated skill-based management

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
	project matching	algorithms	ancestral	stable matchings		
Chiarandini et al., 2019	Fair and satisfactory project allocation	MILP, fairness and stability models	Real institutional data, optimization solver	Fair and stable allocations can be obtained efficiently for their setting	Focuses on allocation optimization	Does not cover complete project lifecycle
Manlove et al., 2022	Maximum stable project allocation	Approximation + Integer Programming	Random + real datasets	Compared approximation solutions with optimal solutions	Computational complexity increases with constraints	No monitoring/evaluation platform
Hoang et al., 2025	Efficient student-project matching	Two-heuristic algorithm	Algorithmic instances	Provides an efficient heuristic approach to stable matching	Mainly allocation focused	Limited personalization beyond preferences
Dynamic Mat	Intelligent mat	Reinforcement	Student profiles,	Supports dynamic project matching	Focus mainly on	Does not integrate monitoring,

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Ching Strategy, 2026	Ching based on interests and skills	Learning	Interests, skills, project metadata	based on student characteristics	matching	evaluation, feedback and management

4. Identify the **limitations and challenges** of the existing research approaches.

Parameter	Stable Matching	MILP	Heuristic	Reinforcement Learning
Preference handling	High	High	High	High
Capacity handling	High	High	High	Can be high
Fairness	Moderate	High	Depends on design	Can be optimized
Computational efficiency	High for suitable cases	Can become expensive	High	Training-dependent
Student skills	Limited	Can be added	Can be added	Naturally suitable
Dynamic adaptation	Low	Low/Moderate	Moderate	High
Personalization	Moderate	High if modeled	Moderate	High potential
Project monitoring	×	×	×	×
Evaluation management	×	×	×	×

Parameter	Stable Matching	MILP	Heuristic	Reinforcement Learning
Faculty feedback	×	×	×	×
Complete academic platform	×	×	×	×

5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.

1. Focus on Allocation Only

Most studies concentrate on the **student-project matching problem** rather than the complete project lifecycle.

2. Limited Student Profile Information

Traditional approaches mainly use preference rankings. They may not sufficiently consider:

- Programming skills
- Technical knowledge
- Previous project experience
- Domain interests
- Learning goals

3. Static Allocation

Once a project is assigned, traditional systems generally do not continuously reconsider whether the project remains suitable for the student.

4. Lack of Monitoring

Most allocation algorithms do not provide facilities for:

- Weekly progress
- Project milestones

- Deadlines
- Faculty monitoring

5. Limited Evaluation Integration

Project allocation research generally does not integrate:

- Internal evaluation
- External evaluation
- Marks
- Faculty feedback

6. Lack of Centralized Management

Students, faculty, coordinators, HODs, and management are not always connected through one integrated platform.

6. Discuss the **possible improvements or future research directions** based on the identified research gap.

Main Research Gap

The major research gap identified from the literature is:

Existing research mainly concentrates on optimizing student-project allocation, while there is limited integration of interest- and skill-based allocation with continuous project monitoring, faculty guidance, evaluation, feedback, deadlines, and centralized academic management in a single system.

Existing Approach

Student Preferences



Project Matching



Project Allocation

Proposed Approach

Student Profile



Interests + Technical Skills



Intelligent Project Recommendation



Project Allocation



Faculty Guide Assignment



Progress Monitoring



Deadlines + Milestones



Faculty Feedback



Internal/External Evaluation



Marks + Reports

7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.

1. Skill-Based Project Recommendation

Instead of considering only student preferences, the system can calculate project suitability using:

- Student interests
- Programming languages

- Technical skills
- Previous experience
- Project requirements

2. AI-Based Recommendation

An AI recommendation model could generate a suitability score:

Student Skills

+

Student Interests

+

Project Requirements

↓

AI Recommendation Model

↓

Project Suitability Score

↓

Recommended Projects

3. Dynamic Project Allocation

The system could update recommendations when a student's skills or interests change.

4. Progress-Based Monitoring

The system can track:

- Weekly progress

- Milestones
- Deadlines
- Pending tasks
- Faculty comments

5. Evaluation Analytics

The system can analyze evaluation marks and identify students or projects requiring additional support.

6. Fairness-Aware Allocation

The allocation algorithm could consider both **student satisfaction and fairness**, building on the fairness-oriented research of Chiarandini et al.

7. Future AI/RL Integration

The 2026 reinforcement-learning research provides a direction for making allocation dynamic and personalized based on student interests, skills, and interactions.

8. Provide proper **citations and references** using a consistent referencing style.

[1] D. F. Manlove and G. O'Malley, "Student-Project Allocation with Preferences over Projects," *Journal of Discrete Algorithms*, vol. 6, no. 4, pp. 553–560, 2008.

[2] D. J. Abraham, R. W. Irving, and D. F. Manlove, "Two Algorithms for the Student-Project Allocation Problem," *Journal of Discrete Algorithms*, vol. 5, no. 1, pp. 73–90, 2007.

[3] M. Chiarandini, R. Fagerberg, and S. Gualandi, "Handling Preferences in Student-Project Allocation," *Annals of Operations Research*, vol. 275, pp. 39–78, 2019.

[4] D. Manlove, D. Milne, and S. Olaosebikan, "Student-Project Allocation with Preferences over Projects: Algorithmic and Experimental Results," *Discrete Applied Mathematics*, vol. 308, pp. 220–234, 2022.

[5] H. H. Viet, N. T. Uyen, S. T. Cao, and L. G. Nguyen, “An Efficient Two-Heuristic Algorithm for the Student-Project Allocation with Preferences over Projects,” *Journal of Intelligent & Fuzzy Systems*, vol. 49, no. 1, 2025.

[6] “Dynamic Matching Strategy for College Students’ Innovative Training Projects Based on Reinforcement Learning Optimization,” *Discover Artificial Intelligence*, vol. 6, article 375, 2026.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing

Future Directions & Relevance(15 %)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation , Citations & References(15 %)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25